

PROBABILITY III

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Chapter 1

REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even n steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

Definition 1.1. A *discrete probability space* is a pair (Ω, q) where Ω is a non-empty finite or countably infinite set called the sample space, and $q : \Omega \rightarrow [0, 1]$ is a function satisfying $\sum_{\omega \in \Omega} q(\omega) = 1$. Any subset of Ω is termed an event, and the probability of an event $A \subseteq \Omega$ is defined as $\mathbb{P}(A) = \sum_{\omega \in A} q(\omega)$.

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \leq i < n\}, \quad (1.1)$$

and the mapping q to be $q(\mathbf{x}) = 2^{-n}$ for all $\mathbf{x} \in \Omega$. Then, the event that the ant returns to the origin after n steps is given by $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$, and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x} \in A} q(\mathbf{x}) = 2^{-n} |A| = 2^{-n} \binom{n}{n/2}. \quad (1.2)$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \quad (1.3)$$

This Ω is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of Ω . As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval $[0, 1]$. If we attempt to assign a positive probability to each element of $[0, 1]$, we would have to assign a probability of 0 to each element, since there are uncountably many elements in $[0, 1]$. However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from $[0, 1]$ is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

Definition 1.2. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set called the sample space, $\mathcal{F}$ is the collection of all subsets of Ω ($\sim 2^\Omega$), and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a function satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity, that is, for any countable collection of pairwise disjoint sets $A_1, A_2, \dots \in \mathcal{F}$,

we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \quad (1.4)$$

Using this definition, we look to define a $\mathbb{P}$ on subsets of $[0, 1]$ that satisfies the above properties, along with the property that $\mathbb{P}([a, b]) = b - a$. Thus, we have $\mathbb{P}([0, 1]) = 1$. If we have two disjoint intervals $I_1, I_2 \subseteq [0, 1]$, then $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$. This is a natural property to expect from a probability measure. What about a general set $S \subseteq [0, 1]$? One can observe that if we have a sequence of intervals $I_1, I_2, \dots$ such that $S \subseteq \bigcup_{k=1}^{\infty} I_k$, then we would have $\mathbb{P}(S) \leq \sum_{k=1}^{\infty} \mathbb{P}(I_k)$. This is because the probability of S should be less than or equal to the probability of any set that contains S . Thus, we can define

$$\mathbb{P}_*(S) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| \mid S \subseteq \bigcup_{k=1}^{\infty} I_k, I_k \subseteq [0, 1] \right\}. \quad (1.5)$$

We have not verified that this is a valid definition of a probability measure, hence the subscript $*$. Via this, let us try computing $\mathbb{P}_*(\mathbb{Q} \cap [0, 1])$. Enumerate the rationals in $[0, 1]$ as $\{q_1, q_2, \dots\}$. For any $\epsilon > 0$, we can choose intervals $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0, 1]$ for each $k \in \mathbb{N}$. Note that $\bigcup_{k=1}^{\infty} I_k$ contains all the rationals in $[0, 1]$, and $\sum_{k=1}^{\infty} |I_k| \leq \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$. Since ϵ was arbitrary, we have $\mathbb{P}_*(\mathbb{Q} \cap [0, 1]) = 0$. This is consistent with our intuition that the probability of choosing a rational number from $[0, 1]$ is 0. So, $\mathbb{P}_*$ seems to be a reasonable definition of a probability measure on $([0, 1], 2^{[0, 1]})$. However, we will show later that there exists a set $\mathcal{A} \subseteq [0, 1]$ such that $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0, 1] \setminus \mathcal{A}) = 1$, contradicting the property that $\mathbb{P}_*([0, 1]) = 1$. We state the following theorem without proof.

Theorem 1.3. *There is no probability measure $\mathbb{P}$ on $([0, 1], 2^{[0, 1]})$ such that $\mathbb{P}([a, b]) = b - a$ for all $[a, b] \subseteq [0, 1]$.*

If we remove the requirement that the probability measure of a subinterval $[a, b]$ is equal to its length, then we can define a probability measure on $([0, 1], 2^{[0, 1]})$. For example, we can define $\mathbb{P}(S) = 1$ if $1/2 \in S$ and $\mathbb{P}(S) = 0$ otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval $[a, b]$ is equal to its length, we can instead restrict the collection of subsets of $[0, 1]$ on which we define the probability measure.

Definition 1.4. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω satisfying the following properties:
 - $\Omega \in \mathcal{F}$,
 - if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
 - if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a function satisfying:
 - $\mathbb{P}(\Omega) = 1$,
 - if $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Such an $\mathcal{F}$ is termed a σ -algebra on Ω , and such a $\mathbb{P}$ is termed a *probability measure* on $(\Omega, \mathcal{F})$.

1.1 σ -algebras

Example 1.5. Given Ω , trivial possible choices for $\mathcal{F}$ are $\{\emptyset, \Omega\}$ and 2^Ω . A more interesting example is the following: suppose Ω is uncountable. Then $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$ is a σ -algebra on Ω .

Let $S \subseteq 2^\Omega$ be a collection of subsets of Ω . We claim that there exists a notion of the smallest σ -algebra containing S . This is called the σ -algebra generated by S , and is denoted by $\sigma(S)$.

Lemma 1.6. *For any collection $S \subseteq 2^\Omega$, there exists a unique smallest σ -algebra containing S satisfying*

$$\sigma(S) = \bigcap \{ \mathcal{F} \subseteq 2^\Omega \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F} \}. \quad (1.6)$$

Proof. It is easy to check that $\sigma(S)$ is indeed a σ -algebra containing S . If $\mathcal{F}$ is any σ -algebra containing S , then $\sigma(S) \subseteq \mathcal{F}$. Thus, $\sigma(S)$ is the unique smallest σ -algebra containing S . ■

Definition 1.7. Let (X, d) be a metric space (or a topological space), and let $\mathcal{S} \subseteq 2^X$ be the collection of all open sets in X . Then, the σ -algebra $\sigma(\mathcal{S})$ is called the *Borel σ -algebra* on X , and its elements are called *Borel sets*.

We will show that there exists a probability measure on $([0, 1], \mathcal{B})$, where $\mathcal{B}$ is the Borel σ -algebra on $[0, 1]$, which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let $\Omega = \{ \omega = (\omega_1, \omega_2, \dots) \mid \omega_i \in \{0, 1\} \}$, where $\omega_i = 0$ corresponds to a step to the left and $\omega_i = 1$ corresponds to a step to the right. If we let $A = \{0, 1\}$, then $\Omega = A^\mathbb{N}$; consider the discrete topology on A , and the product topology on Ω . If we let

$$\mathcal{S} = \{ U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots \mid U_i = \{0\} \text{ or } \{1\} \} \quad (1.7)$$

then one can show that the σ -algebra generated by $\mathcal{S}$ is the Borel σ -algebra on Ω . We can set $\mathcal{F} = \sigma(\mathcal{S})$, and define a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\mathbb{P}(U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots) = 2^{-k}$ for all $U_i = \{0\}$ or $\{1\}$.

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Example 1.8. Let (X, d) be a separable metric space (that is, there exists a countable dense subset). The Borel σ -algebra on X is then

$$\mathcal{B}(X) = \sigma(\{ B(p, r) \}_{p \in X, r > 0}). \quad (1.8)$$

Note that the usual containment is $\sigma(\{ B(p, r) \}_{p \in X, r > 0}) \subseteq \mathcal{B}(X)$ for a metric space X . So, it suffices to show that if $V \in \tau$, the topology on X , then $V \in \sigma(\{ B(p, r) \}_{p \in X, r > 0})$. Let Q be a countable dense subset of X . Then, for any $V \in \tau$, we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \quad (1.9)$$

where $q_n \in Q \cap V$ and $r_n > 0$ (where r_n is the largest possible radius such that $B(q_n, r_n) \subseteq V$). Since $\sigma(\{ B(p, r) \}_{p \in X, r > 0})$ is a σ -algebra, we have $V \in \sigma(\{ B(p, r) \}_{p \in X, r > 0})$. Thus, we have $\mathcal{B}(X) = \sigma(\{ B(p, r) \}_{p \in X, r > 0})$.

Example 1.9. Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space.

- $\mathcal{F}$ is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a σ -algebra. If $A, B \in \mathcal{F}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$. Finally, $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$.
- Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A_1, A_2, \dots \in \mathcal{F}$. Define

$$\limsup_{n \rightarrow \infty} A_n := \{ x \in \Omega \mid x \text{ belongs to infinitely many } A_n \} \quad (1.10)$$

and

$$\liminf_{n \rightarrow \infty} A_n := \{ x \in \Omega \mid x \text{ belongs to all but finitely many } A_n \}. \quad (1.11)$$

Then, both the $\limsup$ and $\liminf$ belong in $\mathcal{F}$. To see this, simply note that

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k. \quad (1.12)$$

- One may show that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(\bigcup A_n) \leq \sum \mathbb{P}(A_n)$ for any countable collection of sets $A_1, A_2, \dots \in \mathcal{F}$.
- For a nested sequence of sets $A_1 \subseteq A_2 \subseteq \dots$ in $\mathcal{F}$, let $A = \bigcup_{n=1}^{\infty} A_n$ (this is usually denoted by $A_n \uparrow A$). Then, $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$. To this end, let $B_n = A_n \setminus A_{n-1}$ for $n > 1$, and $B_1 = A_1$. Then $A = \bigcup_{n=1}^{\infty} B_n$. Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=2}^N \mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N). \quad (1.13)$$

1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of Ω besides the Borel σ -algebra.

1. π -system: a collection of subsets $S \subseteq \Omega$ such that $A, B \in S$ implies $A \cap B \in S$.
2. λ -system: a collection Λ of subsets of Ω such that $\Omega \in \Lambda$, if $A, B \in \Lambda$ with $A \subseteq B$, then $B \setminus A \in \Lambda$, and if $A_1, A_2, \dots \in \Lambda$, then $\bigcup_{n=1}^{\infty} A_n \in \Lambda$.
3. algebra: a collection $\mathcal{A}$ of subsets of Ω such that $\Omega \in \mathcal{A}$, if $A \in \mathcal{A}$, then $\Omega \setminus A \in \mathcal{A}$, and if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

We had defined $\omega(S)$, the σ -algebra generated by a collection of subsets $S \subseteq 2^\Omega$. We can similarly define $\pi(S)$, the π -system generated by S , $\lambda(S)$, the λ -system generated by S , and $\mathcal{A}(S)$, the algebra generated by S .

Theorem 1.10 (*Sierpiński-Dynkin π - λ Theorem*). *Let $S, \mathcal{F} \subseteq 2^\Omega$ be collections of subsets of Ω . Then,*

1. $\mathcal{F}$ is a σ -algebra if and only if $\mathcal{F}$ is both a π -system and a λ -system.
2. If S is a π -system, then $\lambda(S) = \sigma(S)$.

Proof. 1. If $\mathcal{F}$ is a σ -algebra, then it is trivially both a π -system and a λ -system. Conversely, if $\mathcal{F}$ is both a π -system and a λ -system, we need to verify the axioms of a σ -algebra. We have $\Omega \in \mathcal{F}$ since $\mathcal{F}$ is a λ -system. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$ since $\mathcal{F}$ is a λ -system. Finally, if $A_1, A_2, \dots \in \mathcal{F}$, then we can write $B_n = \bigcup_{i=1}^n A_i \in \mathcal{F}$ since $A_i^c \in \mathcal{F}$ and $B_n^c = \bigcap_{i=1}^n A_i^c \in \mathcal{F}$ since $\mathcal{F}$ is a π -system. Then, $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$, and since $\mathcal{F}$ is a λ -system, we have $B \in \mathcal{F}$. Thus, $\mathcal{F}$ is a σ -algebra.

2. Let S be a π -system. Since $\lambda(S) \subseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π -system. Fix $A \in S$, and define $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$. Note that $S \subseteq \mathcal{C}_A$ because S is a π -system. We show that $\mathcal{C}_A$ is a λ -system. We have $\Omega \in \mathcal{C}_A$ since $A \cap \Omega = A \in \lambda(S)$. If $B_1, B_2 \in \mathcal{C}_A$ with $B_1 \subseteq B_2$, then $A \cap B_1 \subseteq A \cap B_2$, and since $A \cap B_2 \in \lambda(S)$, we have $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$, and thus $B_2 \setminus B_1 \in \mathcal{C}_A$. Finally, if $B_1, B_2, \dots$ are in $\mathcal{C}_A$, then $A \cap B_n \in \lambda(S)$ for all n , and thus $\bigcup (A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$, and hence $\bigcup B_n \in \mathcal{C}_A$. Thus, $\mathcal{C}_A$ is a λ -system. Since $\lambda(S)$ is the smallest λ -system containing S , we have $\lambda(S) = \mathcal{C}_A$, and hence for any $B, C \in S$, we have $B, C \in S$ implies $B \cap C = A \cap C \in S$. Thus, $\lambda(S)$ is a π -system. ■

Lemma 1.11. *Let S be a π -system, and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$. Then, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(S)$.*

Proof. Let $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$. Then, $S \subseteq \mathcal{C}$, and we show that $\mathcal{C}$ is a λ -system. We have $\Omega \in \mathcal{C}$ since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$. If $B_1, B_2 \in \mathcal{C}$ with $B_1 \subseteq B_2$, then $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$, and hence $B_2 \setminus B_1 \in \mathcal{C}$. Finally, if $B_1, B_2, \dots$ are in $\mathcal{C}$, then $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$, and since $\mathbb{Q}(A) = \lim_{n \rightarrow \infty} \mathbb{Q}(A_n)$, we have $\mathbb{P}(A) = \mathbb{Q}(A)$. ■

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